



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1st August
2026

Course Number : CE F423
Course Title : Green Buildings and Energy Conservation
Instructor-in-charge : RAJIV GUPTA

Course description: Climate zones and sun path diagram, thermal comfort, heat flow through building materials, energy efficient building design factors like site planning, plan form and orientation, construction techniques, materials and finishes, natural day lighting and ventilation strategies, thermal performance of building elements, simple techniques to recycle and reuse water, harvest rainwater, green building rating system, case studies of traditional architecture and contemporary buildings, building design using AUTOCAD/ GIS/ AI

Scope and Objective: The course introduces green architecture and intends to equip students with technical knowledge of energy-efficient green building. The course covers various aspects of green architecture like climatology, passive solar architecture, water management. The course will also guide students, through projects, to apply concepts and ideas for the design of a green building.

Course learning outcomes:

- By the end of the course, the students should have
1. General knowledge of the field of Green Buildings
 2. They should be able to recognize when and how to use PSA techniques to make the structure sustainable in terms of energy/ carbon footprint
 3. The students should also be able to evaluate new techniques they encounter.
 4. Conservation of water
 5. Students will take up a group project/ term paper to study in detail the theoretical details of an energy efficient system and implement that during the semester
 6. Application of AI, GIS, BIM and use of software to design the energy efficient systems

Text books and references:

- TB1: Krishnan A., Baker N., Yannas S. and Szokolay S. (Ed.). Climate responsive architecture, a design handbook for energy efficient buildings. Tata McGraw-Hill Publishing Company: New Delhi. 2001
- TB2. The Energy and Resources Institute and ICAEN (Institut Catala d'Energia). Sustainable building design manual (Volume 2). The Energy and Resources Institute: New Delhi. 2004.
- R1. Olgyay V. and Olgyay A. Design with climate; bioclimatic approach to architectural regionalism. University Press: New Jersey. 1963.
- R2. Duffie J. and Beckman W. Solar engineering of thermal processes. Second edition. John Wiley & Sons: New York. 1991.
- R3. Bureau of Indian Standards. SP:41, Handbook on functional requirements of buildings (other than industrial buildings). First reprint. Bureau of Indian Standards: New Delhi. 1995.
- R4. Indian Green Building Council. LEED-India, Green building rating system, abridged reference guide for new construction and major renovations (LEED India NC), version 1.0. Indian Green Building Council: Hyderabad.

Method of conduct of the course:

The traditional method of teaching by chalk and board has been highly regarded for many educational levels. The chalk-and-board teaching system has been appreciated for creating an opportunity for direct and personal communication between teachers and students to engage with each other face to face. Approaches such as blended learning, collaborative strategies, and experiential learning promote engagement, critical thinking, and lifelong learning.

Course Plan: weekly schedule for 16 weeks with [3] lectures per week

Week no.	Modxule	Lect. no.	Reference	Learning Outcome
1.	Overview of GBEC, Climate zones and sun path diagram	1	Handout	Methodology, Evaluation Scheme
		2	T1, R1, R3	Climate
2.		3-4	Class notes	Architecture: Passive Solar
2.		5	T1	Sun path and beam radiation

3.	Thermal comfort, heat flow through building materials, energy efficient building design	6	T2	Methods to enhance the Thermal Comfort
		7-8	T2	Materials with different properties and their usage
4.		9 -11	Class notes	Application and problem solving
5.	Site planning, plan form and orientation, construction, techniques, materials and finishes	12-14	T2	Selection of site as per climatic conditions
6 - 7		15-16	T2	Plan form and orientation as per sun path
		17-19	Class notes	Applications of AI & GIS and problem solving
8-10	Natural day lighting and ventilation strategies, thermal performance of building elements	20-21	T2	Carbon footprint
		22-24	T2	LCCA
		25-28	Class notes	Applications of software: AutoCAD, ARCGIS and problem solving
11-12	Efficient water management in buildings	29	T1	Methods of water conservation: Fixtures, rainwater harvesting
		30-32	T1	Reuse of water
13-15	Green building rating system	33-35	T2	Requirements
		36-38	T2	GRIHA, LEED
		39-41	Self	Application, Implementation of Civil Engineering projects
16	Vernacular architecture and its use in contemporary design, Case studies of contemporary green building	42	Self	Vastu Sastra and application in modern buildings

Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Problem solving	20%	30 Class + 20	Once in a week	Buffer 20%
Project	30%	Continuous	Title, Group, Methodology, Ms, Final	All components are must
Webinar	15%	90 min *4	Once in a month	3/4
Comprehensive	35%	180 min	13/12 FN	OB

Criterion for NC:

1. As per academic regulation
2. Final Project Marks will be awarded only if the student completes project as per the course requirement.
3. Missing any component completely will lead to NC

Chamber consultation hours: TH- 9th Hour

Makeup Policy: Prior permission is must.